

HARVARD MATHEMATICS END OF YEAR

2024 BY MR. A. N (0779552488) 3HRS

SECTION A: ANSWER ALL

- A1** Express
- (a) $\frac{12}{25}$ as a decimal fraction, [1]
- (b) $\frac{2}{5}$ as a percentage, [1]
- (c) 0.0375 as a fraction in its lowest terms. [1]
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- 4004/1 J2019 Q1*

- A2** Evaluate, giving your answer in its lowest terms,
- (a) $\frac{4}{5} - \frac{9}{10}$, [1]
- (b) $\frac{7}{15} \times \frac{3}{8} \div \frac{7}{8}$, [2]
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- 4030/1 N2017 Q7*

- A3** Express $\frac{3}{x-2} - \frac{4}{x+1}$ as a simple fraction in its simplest form. [3]
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- 4008/4028/1 N2010 Q12*

- A4** (i) Express $\frac{x}{3} + \frac{x-4}{5}$ as a single fraction in its simplest form.
- (ii) Hence or otherwise solve the equation $\frac{x}{3} + \frac{x-4}{5} = 4$. [4]
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- 4008/4028/2 J2011 Q2(b)*

- A5** Solve the equation $x(x+1) + 2x(x-1) = 3(x^2-1)$. [3]
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- 4008/2 N2011 Q3(a)*

- A6** Find the equation of a straight line which passes through P(3; -4) and Q(-1; 2). [2]
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- 4008/2 J2012 Q6(a)*

- A7** Solve the equation $3 - (2n - 5) = 32$. [2]
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- 4008/1 J2015 Q22(a)*

- A8** Simplify $11011_2 + 243_5$, giving your answer in base 5. [3]
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- 4004/2 J2020 Q1(e)*

- A9a.** i) Solve the following inequalities giving the answer in the form $a \leq x < b$ where a and b are constants to be found: $5x - 13 \leq x - 6 < 9 + 4x$. [3]
- ii) Illustrate the solution on a number line. [1]
- iii) Write down the smallest integer value of x that satisfies the inequality. [1]
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- 4004/2 J2019 Q7(a)*

- A10a** (i) Solve the inequality $2(x-3) < 7$. [2]
- (ii) Write down the largest perfect square that satisfies the inequality $2(x-3) < 7$. [1]
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- 4030/1 J2018 Q20(a)*
- (b) Find the Highest Common Factor (H.C.F) of $2^3 \times 3^2 \times 5 \times 7^4$, $2^3 \times 3^3 \times 5^2 \times 7^2$, $2^4 \times 3 \times 5 \times 7^3$, leaving the answer in index form. [2]
- (c) Find the Lowest Common Multiple (L.C.M) of $3x^2y$, $5x^3y^2$ and $8xy^3$. [2]
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- 4030/2 J2017 Q1*

- A11a** (i) Factorise completely $x^2 - 6x$. [1]
- (ii) Factorise completely $(m-1)^2 - 9$. [2]
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- 4030/2 N2017 Q1(c)*

- b.** Remove the brackets and simplify $(a-b)(2a-b)$. [2]
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- 4030/2 N2016 Q1(a)*

- A12a** Expand $(2a-b)(1+c)$. [2]
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- 4008/1 J2014 Q4(a)*

- b.** Remove the brackets and simplify $(a+2)(a-3) - 3(a-5)$. [3]
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- 4028/2 J2014 Q1(b)*

- c.** Evaluate (a) $(-3)^0$ [1]

SECTION B

CHOOSE ANY 3 QUESTIONS

B13a

The points A(6; 2) and B(8; 5) lie on a straight line.

Find the

- gradient of the line AB, [1]
- equation of the line AB, giving the answer in the form $y = mx + c$. [2]

4004/1 N2018 Q11

b

The following is an incomplete table of values for the function $y = x^2 - 4x$.

x	-2	-1	0	1	2	3	4	5	6
y	12	5	0	p	-4	-3	0	q	12

- Find the values of (i) p , [1]
(ii) q . [1]

Answer parts (b) and (c) of the question on a grid.

- (i) Draw the graph of $y = x^2 - 4x$ on the grid using a scale of 2 cm to 1 unit on the x -axis and 2 cm to 2 units on the y -axis. [4]

4004/2 J2019 Q8

B14

It is given that the universal set, ξ , has subsets P , S , and M such that,

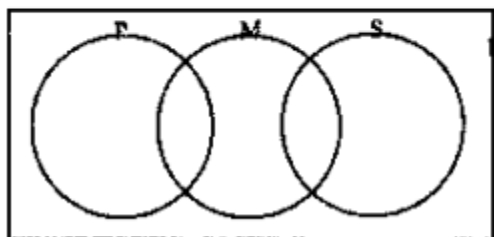
$$\xi = \{1; 2; 3; 4; 5; 6; 7; 8; 9\},$$

$$P = \{\text{prime numbers}\},$$

$$S = \{\text{perfect square numbers}\},$$

$$M = \{\text{multiples of 3}\}.$$

- List all elements of set P . [2]
- Write down $n(P \cap S \cap M)$. [1]
- Complete the Venn diagram by inserting elements in the correct regions. [3]



4004/1 N2018 Q24

Find the value of $(y + z)^x$, given that $x = 2$,

$$y = -2$$

$$\text{and } z = 5.$$

[2]

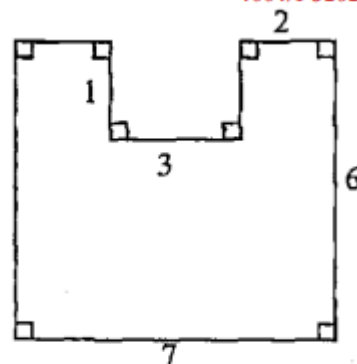
4030/1 N2016 Q8(b)

B15a.

i

A straight line has gradient -1 and passes through the point (3; 0). Find the equation of the line in the form $y = mx + c$. [2]

4004/1 J2020 Q24(a)



Calculate the

- perimeter of the floor, [2]
- area of the floor, [2]

It is given that $\xi = \{1; 2; 3; 4; 5; 6; 7; 8; 9; 10\}$, where H is a set of prime numbers and K is a set of odd numbers.

15.iv.

- List the elements of set H . [1]
- Find $n(H \cup K)$. [2]

4008/1 N2014 Q3

A16.i

The matrix $A = \begin{pmatrix} -2 & 14 \\ 2 & x \end{pmatrix}$ and $|A| = -2$.

- Find the value of x . [2]
- Hence find A^{-1} . [2]

4030/1 J2016 Q21

It is given that

$$3 \begin{pmatrix} p & -1 \\ 0 & 4 \end{pmatrix} - \begin{pmatrix} 7 & q \\ -2 & 2r \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 16 & 8 \\ 4 & -12 \end{pmatrix}$$

.ii.

- Find the value of
- p [2]
 - q [2]
 - r [2]

4030/1 J2018 Q25

70 MARKS